A Synopsis Report
on

Impact of Database Management Systems on Organizational Performance in an Industry

Submitted in partial fulfillment
for the award of the degree of

Bachelor of Technology
in
Computer Science Engineering (Artificial Intelligence & Data Science)

Submitted By

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Project Credentials

Project title	Impact of Database Management Systems on Organizational Performance in an Industry
Project Domain/Area	Database Management Systems, Industrial Information Systems
Group members	**1.** Nirmal 2822581 **2.** kartik 2822530 **3.** Sushant 2822532 **4.** Shubham Pangal 2822512
Group Id *(to be allotted by project coordinator)*	G19
Supervisor's Name	Dr. B.K Verma
Supervisor's Designation	HOD CSE-AIDS

Supervisor's Consent

The synopsis of final year project work titled…… by the students' group id.…G19…… has been written with my consent and every section of this synopsis report is reflecting the work to be carried out by the group.	*(Signature of supervisor with date)*

Department Project Evaluation Committee (DPEC) Remarks

The project is ..... by DPEC. The group is advised to submit progress of the project work in progress presentation-1 to be held on…….

OR

The project is ..... by DPEC. The group is advised to submit the synopsis report again after making changes as suggested by DPEC on …….....

Name & Signature of DPEC member (s) with date

Impact of Database Management Systems on Organizational Performance in an Industry

Recommendation with Engineering Program Outcomes and Program/Department Specific Outcomes

Project Title	Engineering Program Outcomes (POs)	Department Specific Outcomes (DSOs)
Impact of Database Management Systems on Organizational Performance in an Industry	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO11	PSO1, PSO2

Project Categorization

• **Research-based:**
The project involves analytical study of database management systems and their role in improving organizational performance in industries.

• **Application-based:**
The project focuses on real-world industrial applications of database systems.

Project Title	Project Categorization
Impact of Database Management Systems on Organizational Performance in an Industry	Application based

Project Map with SDG Goals

- **Sustainable Development Goals (SDGs)**:
  SDG 8 – Decent Work and Economic Growth: Improved productivity through efficient data management.
  SDG 9 – Industry, Innovation and Infrastructure: Databases form the backbone of industrial information infrastructure.
  SDG 12 – Responsible Consumption and Production: Reduced redundancy and better utilization of data resources.

- **Justification**

  In modern industries, large volumes of data are generated daily from various business processes such as production, sales, inventory, and customer management. Traditional data handling methods are inefficient and error-prone. Database Management Systems (DBMS) provide a structured approach to data storage, retrieval, and security.

  This project is justified by the increasing reliance of industries on databases to improve accuracy, efficiency, and decision-making. Effective database systems enhance organizational performance by reducing redundancy, improving data consistency, and supporting strategic planning.

Introduction

Database Management Systems play a crucial role in today's industrial environment. Industries depend on databases to manage operational data, support business intelligence, and improve decision-making.

The project titled **"Impact of Database Management Systems on Organizational Performance in an Industry"** explores how the use of DBMS improves efficiency, productivity, and performance in industrial organizations. The project also highlights the role of Python in analyzing and managing database-driven applications.

Objectives and Scope

Objectives:

- To understand the role of DBMS in industries
- To analyze the impact of database systems on organizational performance
- To study how databases improve data accuracy and decision-making
- To identify challenges in database implementation

Scope:

The scope of the project includes analysis of database systems used in industries, their benefits, limitations, and their influence on operational efficiency. Python is used for basic database interaction and data analysis.

Methodology

The project follows a descriptive and analytical methodology. Data is collected through literature surveys, case studies, and industrial reports. Python programming is used to interact with databases, analyze data, and observe performance improvements enabled by database systems.

Hardware & Software Requirements

Hardware:

- Minimum 4 GB RAM
- Intel i3 / Ryzen 3 processor or higher
- Personal Computer

Software:

➢ **Programming Language:** Python
➢ **Database:** MySQL / PostgreSQL
➢ **Database Connectivity:** MySQL Connector / Psycopg2
➢ **Data Analysis Library:** Pandas, NumPy

➢ **Visualization Tools:** Matplotlib / Seaborn
➢ **Development Tools:** VS Code, Git, GitHub
➢ **Operating System:** Windows / Linux

Limitations

- Limited access to real industrial data
- Study is based on secondary data sources
- Performance analysis may vary across industries

Future Scope.

- Integration of Big Data and NoSQL databases
- Use of Machine Learning with databases
- Cloud-based industrial database systems
- Advanced data analytics using Python

Conclusion

This project highlights the importance of Database Management Systems in enhancing organizational performance in industries. Proper database implementation leads to improved productivity, better decision-making, reduced errors, and efficient data handling. The use of Python further enhances database interaction and analytical capabilities, making DBMS a vital component of modern industries.

References

1. Elmasri, R., & Navathe, S., *Fundamentals of Database Systems*
2. Silberschatz, A., Korth, H., *Database System Concepts*
3. Research papers on Industrial Database Applications